Using those formulas, we are able to prove the following proposition, but first we introduce two notions. Fix $\ell \in \{1, \dots, n\}$. We say that a permutation $\tau \colon \{1, \dots, 2n\} \to \{1, \dots, 2n\}$ *fixes* $X_\ell = U_\ell$ and $Y_\ell = U_{n+\ell}$ if $\tau(\ell) = \ell$ and $\tau(n+\ell) = n+\ell$. Moreover, we say that it *swaps* $X_\ell = U_\ell$ and $Y_\ell = U_{n+\ell}$ if $\tau(\ell) = n+\ell$ and $\tau(n+\ell) = \ell$. We denote by $\mathcal{P}$ the set of all permutations which either fix or swap X_i and Y_i for all $i = 1, \dots, n$. Since the identity belongs to $\mathcal{P}$, every element in $\mathcal{P}$ is self-inverse, and the composition of two permutations in $\mathcal{P}$ gives an element in $\mathcal{P}$, the set $\mathcal{P}$ is a subgroup of the permutation group.

Proposition 11 *Assume we have equal sample sizes $m = n$ and that we work with the estimator $\widehat{\mathrm{MMD}}^2_{\lambda,\mathsf{b}}$ defined in Equation (6). Then, using a wild bootstrap is equivalent to using permutations which belong to the subgroup $\mathcal{P}$. There is a one-to-one correspondence between these two procedures.*

Proof First, note that for a permutation $\sigma \in \mathcal{P}$, we either have $\sigma(i) = i$, $\sigma(n+i) = n+i$ or $\sigma(i) = n+i$, $\sigma(n+i) = i$ for $i = 1, \dots, n$. Hence, the two sets

$$\{k_\lambda(U_i, U_i), k_\lambda(U_{n+i}, U_{n+i}), k_\lambda(U_i, U_{n+i}), k_\lambda(U_{n+i}, U_i) : i = 1, \dots, n\}$$

and

$$\big\{k_\lambda(U_{\sigma(i)}, U_{\sigma(i)}), k_\lambda(U_{\sigma(n+i)}, U_{\sigma(n+i)}), k_\lambda(U_{\sigma(i)}, U_{\sigma(n+i)}), k_\lambda(U_{\sigma(n+i)}, U_{\sigma(i)}) : i = 1, \dots, n\big\}$$

are equal, we deduce that for $\sigma \in \mathcal{P}$ we have $K_\lambda^{\circ\sigma} = K_\lambda^\circ$. Moreover, since a permutation $\sigma \in \mathcal{P}$ either fixes or swaps X_i and Y_i for $i = 1, \dots, n$, it must be self-inverse, that is $\sigma^{-1} = \sigma$. For the permutation $\sigma \in \mathcal{P}$, we then have

$$\widehat{M}_\lambda^\sigma = \frac{1}{n(n-1)}\, v_\sigma^\top K_\lambda^\circ v_\sigma.$$

for $v_\sigma = (v_{\sigma(1)}, \dots, v_{\sigma(2n)})$ where $v_i = 1$ and $v_{n+i} = -1$ for $i = 1, \dots, n$. We recall that for the wild bootstrap we have n i.i.d. Rademacher random variables $\epsilon \coloneqq (\epsilon_1, \dots, \epsilon_n)$ with values in $\{-1, 1\}^n$ and

$$\widehat{M}_\lambda^\epsilon = \frac{1}{n(n-1)}\, v_\epsilon^\top K_\lambda^\circ v_\epsilon.$$

where $v_\epsilon \coloneqq (\epsilon, -\epsilon) \in \{-1, 1\}^{2n}$.

We need to show that for a given permutation $\sigma \in \mathcal{P}$ there exists some $\epsilon \in \{-1, 1\}^n$ such that $v_\epsilon = v_\sigma$, and that for a given $\epsilon \in \{-1, 1\}^n$ there exists a permutation $\sigma \in \mathcal{P}$ such that $v_\sigma = v_\epsilon$, and that this correspondence is one-to-one.

Suppose we have a permutation $\sigma \in \mathcal{P}$ and let $\epsilon \coloneqq (v_{\sigma(1)}, \dots, v_{\sigma(n)}) \in \{-1, 1\}^n$. We claim that $v_\sigma = v_\epsilon$, that is, that $(v_{\sigma(1)}, \dots, v_{\sigma(2n)}) = (v_{\sigma(1)}, \dots, v_{\sigma(n)}, -v_{\sigma(1)}, \dots, -v_{\sigma(n)})$, so we need to prove that $v_{\sigma(n+i)} = -v_{\sigma(i)}$ for $i = 1, \dots, n$. As $\sigma \in \mathcal{P}$, for $i = 1, \dots, n$, we either have $\sigma(i) = i$ and $\sigma(n+i) = n+i$ in which case

$$v_{\sigma(n+i)} = v_{n+i} = -1 = -v_i = -v_{\sigma(i)},$$

or $\sigma(i) = n+i$ and $\sigma(n+i) = i$ in which case we have

$$v_{\sigma(n+i)} = v_i = 1 = -v_{n+i} = -v_{\sigma(i)}.$$

This proves the first direction.

Now, suppose we are given $\epsilon \coloneqq (\epsilon_1, \dots, \epsilon_n)$ i.i.d. Rademacher random variables. We have $v_\epsilon = (\epsilon, -\epsilon) \in \{-1,1\}^{2n}$ and we need to construct $\sigma \in \mathcal{P}$ such that $v_\sigma = v_\epsilon$, that is, $v_{\sigma(i)} = \epsilon_i$ and $v_{\sigma(n+i)} = -\epsilon_i$ for $i = 1, \dots, n$. We can construct such a permutation $\sigma \in \mathcal{P}$ as follows:

for $i = 1, \dots, n$:
  if $\epsilon_i = 1$ **then let** $\sigma(i) \coloneqq i$ **and** $\sigma(n+i) \coloneqq n+i$ (i.e. σ fixes X_i and Y_i)
    we then have $v_{\sigma(i)} = v_i = 1 = \epsilon_i$ and $v_{\sigma(n+i)} = v_{n+i} = -1 = -\epsilon_i$
  if $\epsilon_i = -1$ **then let** $\sigma(i) \coloneqq n+i$ **and** $\sigma(n+i) \coloneqq i$ (i.e. σ swaps X_i and Y_i)
    we then have $v_{\sigma(i)} = v_{n+i} = -1 = \epsilon_i$ and $v_{\sigma(n+i)} = v_i = 1 = -\epsilon_i$

This proves the second direction.

Our two constructions show that the correspondence is one-to-one. ■

This highlights the relation between those two procedures: using a wild bootstrap is equivalent to using a restricted set of permutations for the estimator $\widehat{\mathrm{MMD}}^2_{\lambda,\mathtt{b}}$.

Appendix C. Efficient implementation of MMDAgg (Algorithm 1)

We discuss how to compute Step 1 of Algorithm 1 efficiently. To compute the $|\Lambda|$ kernel matrices for the Gaussian and Laplace kernels, we compute the matrix of pairwise distances only once. For each $\lambda \in \Lambda$, we compute all the values $\big(\widehat{M}^b_{\lambda,1}\big)_{1\leq b\leq B_1+1}$ and $\big(\widehat{M}^b_{\lambda,2}\big)_{1\leq b\leq B_2}$ together. We compute the sums over the permuted kernel matrices efficiently, in particular we do not want to explicitly permute rows and columns of the kernel matrices as this is computationally expensive.

We start by considering the wild bootstrap case, so we have equal sample sizes $m = n$. We let K_λ denote the kernel matrix $(k_\lambda(U_i, U_j))_{1\leq i,j\leq 2n}$ where $U_i \coloneqq X_i$ and $U_{n+i} \coloneqq Y_i$ for $i = 1, \dots, n$. Note that K_λ is composed of four $(n \times n)$-submatrices, we denote by K°_λ the matrix K_λ with the diagonal entries of those four submatrices set to 0. As explained in Appendix B, for n i.i.d. Rademacher random variables $\epsilon \coloneqq (\epsilon_1, \dots, \epsilon_n)$ with values in $\{-1,1\}^n$, we have

$$\widehat{M}^\epsilon_\lambda = \frac{1}{n(n-1)}\, v_\epsilon^\top K^\circ_\lambda\, v_\epsilon$$

where $v_\epsilon \coloneqq (\epsilon, -\epsilon) \in \{-1,1\}^{2n}$. We want to extend this to be able to compute $\big(\widehat{M}^{\epsilon^{(b)}}_\lambda\big)_{1\leq b\leq B}$ for any $B \in \mathbb{N}\setminus\{0\}$. We can do this by letting R be the $2n \times B$ matrix consisting of stacked vectors $\big(v_{\epsilon^{(b)}}\big)_{1\leq b\leq B}$ and computing

$$\frac{1}{n(n-1)}\mathrm{diag}\Big(R^\top K^\circ_\lambda R\Big)\,.$$

Note that, in general, given $2n \times B$ matrices $A = (a_{i,j})_{\substack{1\leq i\leq 2n\\1\leq j\leq B}}$ and $C = (c_{i,j})_{\substack{1\leq i\leq 2n\\1\leq j\leq B}}$, we have

$$\mathrm{diag}\Big(A^\top C\Big) = \mathrm{diag}\left(\left(\sum_{r=1}^{2n} a_{r,i}c_{r,j}\right)_{\substack{1\leq i\leq B\\1\leq j\leq B}}\right) = \left(\sum_{r=1}^{2n} a_{r,i}c_{r,i}\right)_{1\leq i\leq B} =: \sum_{\text{rows}} A \circ C$$

where $\circ$ denotes the Hadamard product and where $\sum_{\text{rows}}$ takes a matrix as input and outputs a vector which is the sum of the row vectors of the matrix. We deduce that

$$\operatorname{diag}\Big(R^\top K_\lambda^\circ R\Big) = \sum_{\text{rows}} R \circ (K_\lambda^\circ R).$$

We found that this way of computing the values $\big(\widehat{M}_\lambda^{\epsilon^{(b)}}\big)_{1\leq b\leq B}$ is computationally faster than other alternatives. Letting $\mathbb{1}_n$ denote the vector of length n with all entries equal to 1, we can obtain an efficient version for Step 1 of Algorithm 1 using a wild bootstrap as follows.

<u>*Efficient Step 1 of Algorithm 1 using a wild bootstrap:*</u>
generate $n \times (B_1 + B_2 + 1)$ matrix $\widetilde{R}$ of Rademacher random variables
concatenate $\tilde{R}$ and $-\widetilde{R}$ to form the $2n \times (B_1 + B_2 + 1)$ matrix R
replace the $(B_1 + 1)^{\text{th}}$ column of R with the vector $(\mathbb{1}_n, -\mathbb{1}_n)$
for $\lambda \in \Lambda$:
  compute kernel matrix K_λ° with zero diagonals for its four submatrices
  compute $\frac{1}{n(n-1)} \sum_{\text{rows}} R \circ (K_\lambda^\circ R)$ to get $\big(\widehat{M}_{\lambda,1}^1, \dots, \widehat{M}_{\lambda,1}^{B_1+1}, \widehat{M}_{\lambda,2}^1, \dots, \widehat{M}_{\lambda,2}^{B_2}\big)$
  $\big(\widehat{M}_{\lambda,1}^{\bullet 1}, \dots, \widehat{M}_{\lambda,1}^{\bullet B_1+1}\big) = \texttt{sort\_by\_ascending\_order}\big(\widehat{M}_{\lambda,1}^1, \dots, \widehat{M}_{\lambda,1}^{B_1+1}\big)$

Before tackling the case of permutations, we recall the main steps of the strategy used for the wild bootstrap case. As explained in Appendix B, we first noted that

$$\begin{aligned}
\widehat{\mathrm{MMD}}^2_{\lambda,\mathtt{b}}(\mathbb{X}_n, \mathbb{Y}_n) &= \left(K_\lambda^\circ \circ \begin{pmatrix} \mathbb{1}_n\mathbb{1}_n^\top/(n(n-1)) & -\mathbb{1}_n\mathbb{1}_n^\top/(n(n-1)) \\ -\mathbb{1}_n\mathbb{1}_n^\top/(n(n-1)) & \mathbb{1}_n\mathbb{1}_n^\top/(n(n-1)) \end{pmatrix}\right)_+ \\
&= \frac{1}{n(n-1)} \left(K_\lambda^\circ \circ vv^\top\right)_+ \\
&= \frac{1}{n(n-1)}\, v^\top K_\lambda^\circ\, v.
\end{aligned}$$

for $v := (\mathbb{1}_n, -\mathbb{1}_n) \in \mathbb{R}^{2n\times 1}$, where A_+ denotes the sum all the entries of a matrix A. We then observed that it was enough to replace the vector v with $v_\epsilon := (\epsilon, -\epsilon) \in \{-1, 1\}^{2n}$ to obtain

$$\widehat{M}_\lambda^\epsilon = \frac{1}{n(n-1)}\, v_\epsilon^\top K_\lambda^\circ\, v_\epsilon.$$

The whole reasoning was based on the fact that we could rewrite the matrix

$$\begin{pmatrix} \mathbb{1}_n\mathbb{1}_n^\top/(n(n-1)) & -\mathbb{1}_n\mathbb{1}_n^\top/(n(n-1)) \\ -\mathbb{1}_n\mathbb{1}_n^\top/(n(n-1)) & \mathbb{1}_n\mathbb{1}_n^\top/(n(n-1)) \end{pmatrix}$$

as an outer product of vectors.